

Smart Vehicle-Mounted Pothole Detection and Mapping System with Real-Time Database Integration

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Abstract—The Smart Vehicle-Mounted Pothole Detection and Mapping System enhances road safety and maintenance by utilizing the YOLOv4-Tiny algorithm for real-time pothole detection via dash cam footage. Detected potholes are logged with GPS coordinates and transmitted to a Firebase database, which continuously updates a map of road conditions. A mobile app provides drivers with alerts when approaching a pothole within a user-defined radius, aiding in safe driving and helping authorities prioritize road repairs. The system was tested under various road conditions, achieving a mean Average Precision (mAP) of 80.7% for pothole detection. It was evaluated using different pothole sizes and environments, and the YOLOv4-Tiny model showed reliable detection with minimal false positives. The real-time notification feature effectively alerted drivers, while the cloud-based database provided accurate, actionable data for road maintenance. The system demonstrated efficient performance and scalability, making it a practical solution for improving both road safety and infrastructure upkeep.

Keywords— *Pothole Detection, Convolutional Neural Network, You Only Look Once, Mean Average Precision, Real-time Data Processing, Firebase Realtime Database, Mobile Application Development, Object Detection.*

I. INTRODUCTION

Potholes are a persistent and growing problem on roads worldwide, with serious consequences for both vehicle safety and infrastructure. For instance, in a Country like India, potholes cause significant damage to vehicles, from flat tyres to suspension failures, often resulting in costly repairs. According to Nitin Gadkari, the Minister of Road Transport and Highways, a total of 3,564 road accidents occurred in India due to potholes in 2020. These road hazards also pose a major safety risk. From 2013-2016, a total of 38,105 road crashes occurred due to potholes, resulting in 36,421 injuries and 11,386 deaths. Traditional methods of addressing potholes such as periodic manual inspections or reactive repairs after accidents occur are often inefficient. For example, in many rural areas, road maintenance can be delayed for months, leaving drivers unaware of the hazards ahead. In addition, manual surveys and inspections may miss critical locations due to limited coverage and slow data collection processes. In response to these challenges, the Smart Vehicle-Mounted Pothole Detection and Mapping System offers a modern, technology-driven solution. Picture

a car moving down a busy road, its dashcam continuously capturing high-definition images of the road surface. As the car approaches a pothole, the YOLO v4 Tiny algorithm processes the image in real-time, identifying the pothole with high accuracy. This data is immediately logged and the vehicle's precise GPS coordinates are recorded.

The system doesn't just detect potholes—it transforms this information into a valuable resource for all road users. For instance, when a driver approaches a pothole, an interactive map on their mobile app provides a visual representation of the pothole's location and distance. If the vehicle is within a user-defined radius of a pothole (default is 10 meters, but adjustable), the app sends a real-time alert, warning the driver to slow down or take evasive action.

This method isn't hypothetical; While similar systems have been deployed in countries like Japan to monitor and repair roads, our solution differentiates itself by being more accessible and scalable. It works with any vehicle equipped with a dashcam, avoiding the need for specialized fleets. Additionally, real-time driver notifications through a mobile app enhance safety and provide a more personalized user experience.

By combining real-time detection, crowd-sourced data aggregation, and instant driver notifications, the Smart Vehicle-Mounted Pothole Detection and Mapping System presents a scalable solution that enhances road safety, optimizes maintenance efforts, and reduces the risk of vehicle damage.

II. RELATED WORKS

Hanshen Chen et al. (2020) proposed a method using location-aware convolutional neural network (CNN) for pothole detection with 95.2% precision and 92.0% recall. The method outperformed others but struggled with low-resolution images, signaling a potential area for improvement.[1]

Sung-Sik Park et al. (2021) evaluated YOLO models, including YOLOv4, YOLOv4-Tiny, and YOLOv5s, using real-time pothole detection frameworks. While YOLOv4-Tiny showed the best performance with a mean average precision mean Average Precision (mAP) of 78.7%, one challenge was maintaining accuracy in detecting small potholes at long distances or in poor weather conditions. This

challenge highlights the limitations of 2Dimensional (2D) image-based methods, as adverse lighting or shadows can affect detection.[2]

Kshitija Chavan et al. (2022) developed YOLOv4-based system achieved 80-85% accuracy, focusing on developing a user-friendly Graphical user interface (GUI). Expanding detection to other road anomalies was suggested, a critical improvement area as pothole detection algorithms often miss edge cases like road cracks or deformations. Additionally, implementing broader datasets and diverse conditions could improve system robustness.[3]

Nachuan Ma et al. (2022) Ma et al. compared 2D and 3D imaging techniques and noted a significant gap between theoretical developments and practical applications in pothole detection. 3D techniques, while highly accurate in depth and volume measurement, are more expensive and difficult to deploy in real-world environments, especially when potholes are filled with water or debris.[4]

A. Lincy et al. (2023) developed YOLOv7-based detection system achieved an impressive 94.5% accuracy. However, they emphasized that future work must focus on optimizing the model to withstand variations in weather and lighting conditions, such as nighttime driving, where models like Region-Based Convolutional Neural Network (R-CNN) have been shown to perform better.[5]

Jakubec M. et al. (2023) conducted a comparison of CNN-based models concluded that while YOLO models excelled in clear conditions, R-CNNs were better at handling low-visibility situations, such as at night. This highlights the importance of environmental factors in model performance and the need to tailor models to specific driving conditions.[6]

Ömer KAYA et al. (2023) explored the relationship between vehicle speed and detection accuracy using YOLOv7 and YOLOv8 showed that other factors, such as camera angle and sunlight, had more significant impacts than speed itself, particularly in vision-based methods that rely heavily on lighting.[7]

Khan M et al. (2023) emphasized the high accuracy of YOLOv8 in autonomous vehicles but suggested further testing under diverse real-world conditions to improve robustness. Adapting models to different climatic conditions and road types remains a significant challenge [8].

Mohd Omar et al. (2024) conducted research, using YOLO variants within an Intelligent Transport System, demonstrated YOLOv7's 93% precision. However, limited datasets remain a barrier to greater accuracy and generalizability, highlighting the need for larger and more diverse datasets.[9]

Mostafa Hoseini et al. (2024) integrated YOLOv5 with Strong SORT tracking, this cost-effective system allowed frequent monitoring of road conditions, significantly reducing the time required for road maintenance.[10]

Jaroslav Frnda et al. (2024) conducted a comparative study of YOLOv7 and Fast Region-Based Convolutional Neural Network (FRCNN) models under adverse conditions provided valuable insights into model selection based on

specific applications, such as selecting FRCNN for nighttime or poor weather, where lighting and shadows present more significant challenges.[11]

Shaghouri et al. (2021) presented a real-time pothole detection method using deep learning, offering a robust framework for autonomous detection. The model's effectiveness is demonstrated under various road conditions, though it faces challenges with shadow and lighting variations, common in real-time applications.[12]

Darapaneni et al. (2021) presented a paper discusses the application of advanced neural networks for pothole detection, presented at the IEEE Information Technology, Electronics and Mobile Communication Conference (IEMCON). The system achieved notable accuracy but faced limitations in adapting to different environmental conditions.[13]

III. PROPOSED SYSTEM DESIGN

The proposed architecture of the Smart Vehicle-Mounted Pothole Detection and Mapping System, a comprehensive solution aimed at enhancing road safety and optimizing maintenance efforts. This system is built around the integration of YOLOv4-Tiny, a lightweight object detection model, specifically tailored for real-time operations. By leveraging real-time data collection through vehicle-mounted dashcams, the system is capable of processing and detecting potholes instantly, enabling rapid and accurate identification of road anomalies.

The architecture outlines a seamless workflow where the vehicle-mounted dashcam continuously captures images of the road. These images are processed locally by the Raspberry Pi, which is equipped with the YOLOv4-Tiny model for pothole detection. Once a pothole is detected, the system collects critical metadata, such as Global Positioning System (GPS) coordinates, and directly stores it in a Firebase Realtime Database. This information is then made accessible to users via a mobile application, which provides real-time notifications and a visual map to ensure drivers are well-informed about potential hazards ahead.

The system ensures precision in both the identification and localization of potholes by pairing dashcam images with GPS data. The mobile application not only allows users to view these pothole locations on a dynamic map but also provides proximity-based alerts, enhancing the overall safety for drivers by giving them adequate time to react. This integrated approach ensures that road safety is improved through timely notifications, while simultaneously providing maintenance teams with up-to-date data to address and fix the issues more efficiently.

A. Data Collection and Processing

The first step in this process involves gathering video data through dashcams mounted on the windshields of vehicles. These dashcams continuously capture high-resolution road footage as the vehicle travels. The strategic placement of the dashcams ensures a wide field of view, allowing the system to monitor the road for potholes under different environmental and lighting conditions.

The recorded footage is then transmitted to raspberry pi 4 equipped with the YOLO V4-Tiny model. This model, known for its high efficiency and real-time processing capabilities, analyses each frame to detect potential potholes. The YOLO V4-Tiny model uses advanced deep learning algorithms to pinpoint the precise location of potholes, marking them by bounding boxes and extracting both the image of the pothole and the corresponding GPS coordinates from the vehicle's onboard systems.

B. Data Aggregation and Storage

Once a pothole is detected, the system uploads relevant data including the pothole's image and GPS coordinates to Firebase. Firebase serves as the central repository for storing this information, ensuring real-time updates and synchronization across devices. This setup allows for efficient storage of both high-resolution images and geospatial data, enabling accurate and timely pothole mapping for users and authorities.

The system uses Firebase Realtime Database to store the pothole location data, which includes latitude, longitude, and

pothole detection results. When a pothole is detected, the image is uploaded to Firebase Storage, and the corresponding metadata (such as coordinates and detection timestamp) is stored in the Realtime Database. This allows for real-time updates and access to the most current pothole data, which can be accessed by both the system and users' mobile applications for immediate alerts and road condition monitoring.

C. User Interface and Alerts

The mobile application, developed using Flutter, interfaces with the system's Firebase database to monitor and fetch the latest pothole locations and coordinates. It displays this information on an interactive map using OpenStreetMap, allowing users to view nearby potholes in real-time. When a driver approaches within a user-defined range of a detected pothole, the app issues an audible alert, giving the driver enough time to react. The alert also displays an image of the pothole, enhancing road safety by helping drivers avoid potential hazards.

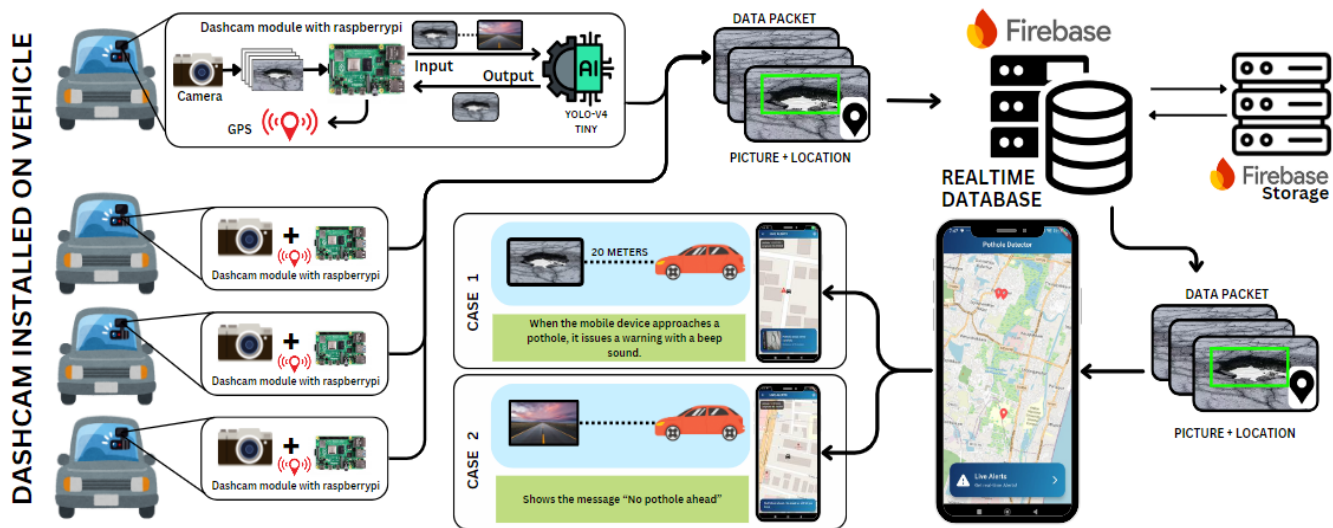


Figure 1: Proposed System Architecture Diagram of Pothole Detection Model.

Figure 1 illustrates the system architecture of the Smart Vehicle-Mounted Pothole Detection and Mapping System. The process begins with the vehicle's dashcam capturing road images, which are processed in real-time by the Raspberry Pi using the YOLOv4-Tiny model for pothole detection. Detected potholes, along with their GPS coordinates, are uploaded to Firebase Storage for image storage, and the corresponding metadata (image URLs and coordinates) is stored in Firebase Realtime Database. The mobile app receives updates from Firebase and generates real-time alerts based on the proximity of detected potholes to the user's current location, ensuring drivers are promptly notified of potential road hazards and enhancing road safety.

IV. PROPOSED SYSTEM IMPLEMENTATION

The proposed system uses real-time image processing and YOLOv4-Tiny to detect potholes. It captures images,

identifies potholes, and provides alerts to drivers through a mobile app, helping with road safety and maintenance.

A. Image Processing

The image processing pipeline begins with continuous image capture from dash cameras mounted on the vehicle. These cameras record real-time video footage as the vehicle moves, which is then segmented into individual frames for analysis. Each captured image is resized to a standard resolution, ensuring uniformity and consistency across the processing stage. Resizing helps optimize computational efficiency, allowing the system to maintain high performance regardless of varying input image sizes or camera resolutions.

The images are subsequently passed through a YOLOv4-Tiny object detection model that has been specifically trained to identify potholes. Before detection, the system

preprocesses each image by normalizing pixel values, which adjusts the image brightness and contrast, helping to improve detection accuracy across different lighting conditions. The YOLO model then extracts essential visual features from the image, such as edges, textures, and shapes, and identifies the presence of potholes. Once a pothole is detected, the model outputs bounding boxes around the pothole's location within the frame, accompanied by confidence scores that help assess the detection's reliability. This data can then be used for further analysis or maintenance actions.

B. YOLOv4-Tiny Architecture

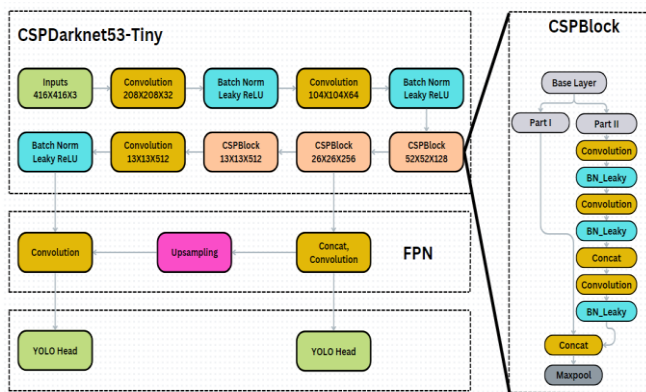


Figure .2: Architecture of YOLOv4-Tiny Detection

Figure .2 shows the architecture of Cross Stage Partial (CSP) Darknet53-Tiny, a lightweight backbone used in YOLO for efficient real-time object detection. The backbone begins with input and convolutional layers to extract key features from the input image, followed by CSP Blocks that split and merge feature channels, improving feature representation while reducing computational overhead. The Feature Pyramid Network (FPN) combines multi-scale features through up-sampling and concatenation, enabling the model to detect objects of varying sizes. The YOLO Head then generates predictions, including bounding boxes and class probabilities, at multiple scales for improved detection accuracy. Within each CSP Block, convolutional, batch normalization, and pooling layers are used to further enhance feature extraction. This compact architecture effectively balances accuracy and speed, making it well-suited for deployment on resource-limited devices.

The YOLOv4-Tiny model, known for its high efficiency and speed in real-time object detection, is employed to detect potholes from the processed frames. The YOLOv4-Tiny model is chosen for real-time pothole detection due to its lightweight architecture and fast image processing, which are essential for real-time applications. With high accuracy and low computational requirements, it is well-suited for edge devices like the Raspberry Pi 4, used in this system. Compared to other models like YOLOv4 and YOLOv5, YOLOv4-Tiny offers a balance between speed and accuracy, achieving strong performance while minimizing latency. This makes it ideal for detecting potholes in moving vehicles, ensuring timely alerts for drivers and efficient data collection for road maintenance.

C. Pothole Detection

Upon detection, the YOLO model outputs a confidence score, which indicates the likelihood of the identified object being a pothole. A predefined threshold determines whether the object is indeed a pothole. When the confidence score surpasses this threshold, the system marks the image as containing a pothole. This detection is performed in real-time as images are continuously processed. The system ensures that the detected potholes are immediately logged into the centralized database along with their GPS coordinates.

D. Continuous Image Capture

The system is designed to capture frames at regular intervals using the vehicle's dash cam. These frames are processed continuously by the YOLO model to detect potholes. Each frame is buffered and processed individually, maintaining a steady flow of data for analysis. The system captures the surrounding road conditions, ensuring that potholes are detected from various angles and under different lighting conditions.

E. YOLO-Based Detection

YOLOv4-Tiny plays a critical role in detecting potholes in real-time. The model processes images captured by the camera, focusing specifically on road sections where potholes are likely to occur. YOLO detects potholes by segmenting each image into grids and identifying objects based on their features. This method allows the system to detect potholes with high accuracy and speed, ensuring real-time detection even when the vehicle is moving at normal driving speeds.

The YOLOv4-Tiny model is good at detecting larger potholes, which are more likely to cause significant damage to vehicles. However, it struggles with smaller potholes, which may not cause immediate damage but can lead to wear on vehicles over time. To improve detection of smaller potholes, the system could be improved by training the model with data focused on small potholes, using better cameras, and applying data augmentation techniques. These improvements would help the system detect smaller potholes more effectively and provide better road condition monitoring.

F. Saving Detected Potholes

Once the YOLO model detects a pothole, the image uploaded to a centralized database for further analysis. The detected pothole is tagged with its GPS coordinates and stored along with the image in a Firebase database. This allows for future access, mapping, and real-time notifications to other drivers who may be approaching the same area. Additionally, the system records the confidence score and other metadata for each detection.

G. Mapping and Notification

After potholes are detected and their data is logged in the database, the system integrates this information with a mobile application. The app fetches the location data of potholes in real-time, displaying them on a map. When a vehicle approaches within user-defined radius of a detected pothole, the app sends an alert to the driver, warning them of the road hazard ahead. The app continuously updates with new

pothole data as it is detected, providing users with up-to-date road conditions and improving overall road safety.

H. Alert System

The system is designed to notify drivers of potential road hazards through a mobile app. When a pothole is detected in real-time, the system stores its location in a database. If another vehicle approaches the same location, an alert is triggered. The alert is delivered as a pop-up notification, providing the driver with information such as the distance to the pothole and a visual of the detected pothole. This early warning system helps drivers take appropriate action to avoid potholes, reducing the risk of vehicle damage and accidents.

I. Circuit Diagram

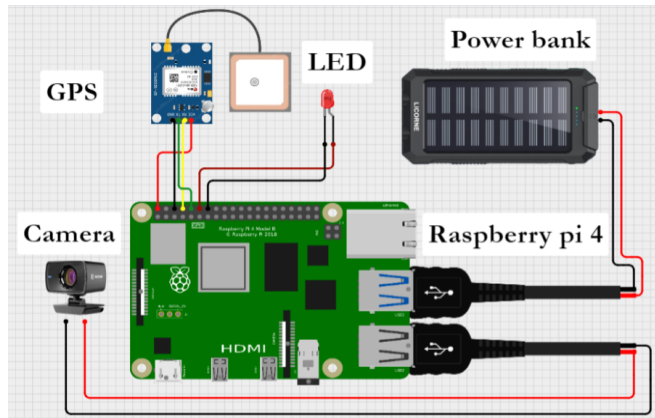


Figure 3: Circuit diagram

Figure 3 shows the hardware setup of the Raspberry Pi 4 is configured with a GPS module connected to Transmit Pin (TX) (Pin 3), Receive Pin (RX) (Pin 5), Voltage at the Common Collector (VCC) (Pin 4), and ground (GND) (Pin 6). A camera is connected via the Camera Serial Interface (CSI), an Light Emitting Diode (LED) is attached to General Purpose Input/Output (GPIO) Pin 18, and a universal serial bus (USB) webcam is connected through the USB port. The system is powered by a portable power bank.



Figure 4: Dashcam prototype

Figure 4 shows the prototype of the dashcam. This system enables real-time detection and mapping of potholes, offering an effective solution to enhance road safety and maintenance by notifying drivers when they approach nearby potholes, improving safety.

J. Mobile Application Integration

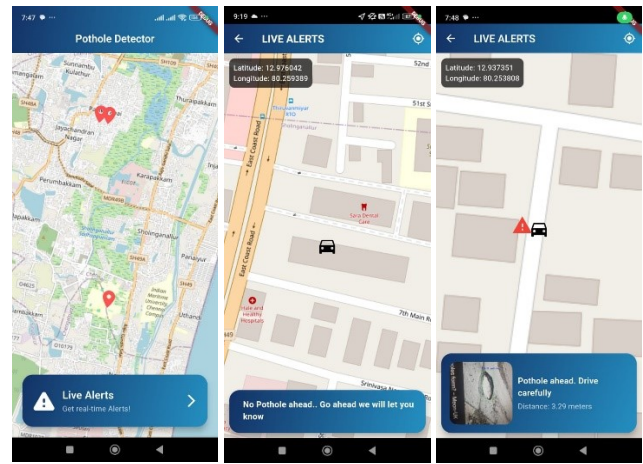


Figure 5: User interface and alerts

Figure 5 illustrates the user interface of the mobile app developed. The app for the pothole detection system has been developed using Flutter, a framework that allows for cross-platform development with a single codebase. The app communicates directly with Firebase, where the pothole detection results from the Raspberry Pi-4 are stored. The app retrieves pothole locations from Firebase and triggers alerts for the user. The user interface (UI) displays pothole alerts on a map using the Google Maps Flutter package to plot locations. Permissions for accessing the camera and location are managed using Flutter's permission_handler package, and background tasks are implemented to continuously monitor for potholes during travel.

The model performs well, but to further enhance accuracy, data augmentation with variations like rain or fog can improve generalization. Fine-tuning on a smaller, diverse pothole dataset can boost detection, while optimizing hyperparameters can improve efficiency. Additionally, refining non-maximum suppression can reduce false positives and improve detection precision.

V. EXPERIMENTAL RESULT

Conducted extensive testing to evaluate the accuracy and efficiency of the integrated "Smart Vehicle-Mounted Pothole Detection and Mapping System." The system leverages the YOLOv4-Tiny model for pothole detection and processes real-time image data. The experiment included a diverse set of road conditions and varying pothole sizes to test the robustness of the system.

The table.1 compares three different models for pothole detection: YOLOv5s, YOLOv4, and YOLOv4-Tiny (proposed). The objective of all models is pothole detection. YOLOv5s achieves a mean Average Precision (mAP) of 74.80%, with its strength lying in its versatility for general tasks. However, it has a limitation of lower mAP compared to other models. YOLOv4, with a higher mAP of 77.70%, is noted for its high accuracy but still lags behind in comparison to YOLOv4-Tiny. The proposed YOLOv4-Tiny, with an mAP of 80.70%, performs well in being lightweight and real-time, though it struggles with detecting small and distant potholes. with trade-offs between accuracy, speed, and capability in handling specific pothole sizes and distances.

Table 1: Comparison of YOLO models

Aspect	Yolo v5s[2]	Yolov4[2]	Yolo v4-Tiny(proposed)
Objective	Pothole detection	Pothole detection	Pothole detection
mAP_0.5	74.80%	77.70%	80.70%
Strengths	General tasks	High accuracy	Lightweight, real-time
Limitations	Lower mAP	Lower than YOLOv4-Tiny	Small, distant potholes

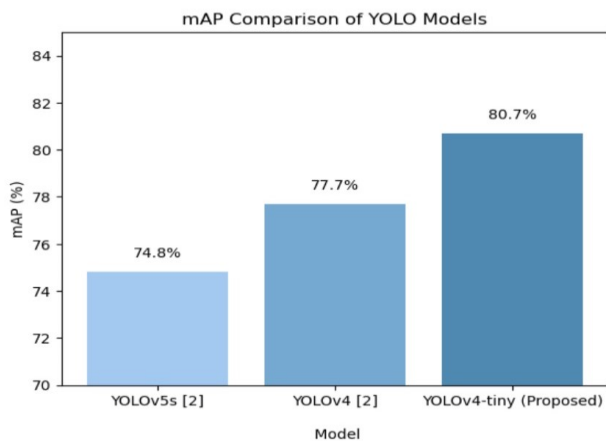


Figure .6: Comparison of YOLO Models by mAP@0.5

Figure .6 compares the mean Average Precision (mAP) of different YOLO models, showing YOLOv4-Tiny (Proposed) achieving the highest accuracy at 80.7%. YOLOv5s and YOLOv4 models scored lower, with 74.8% and 77.7% respectively. The YOLOv4-Tiny-based detection model demonstrated strong performance, successfully identifying potholes under different lighting and weather conditions. The model was trained on a dataset containing various pothole images, allowing it to accurately detect the presence of potholes with minimal false positives.

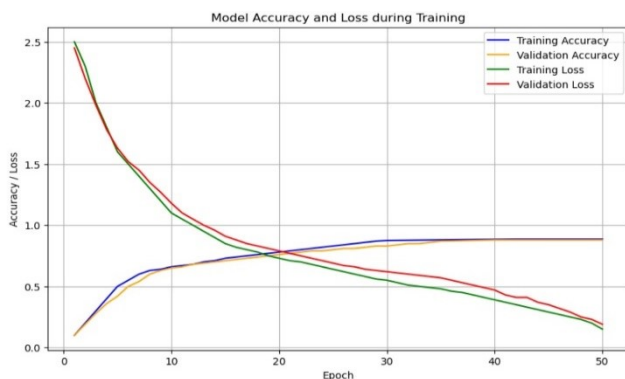


Figure .7: Accuracy and Loss over 50 Epochs

Figure .7 shows the model's performance over 50 epochs. By the final epoch, the training accuracy reached 88.7% and the validation accuracy 87.9%, with minimal difference, indicating good generalization and low overfitting. Training loss decreased from 2.5 to 0.15, while validation loss dropped from 2.45 to 0.19, demonstrating effective learning. The

small gap between training and validation loss further confirms the model's robustness and ability to generalize well.

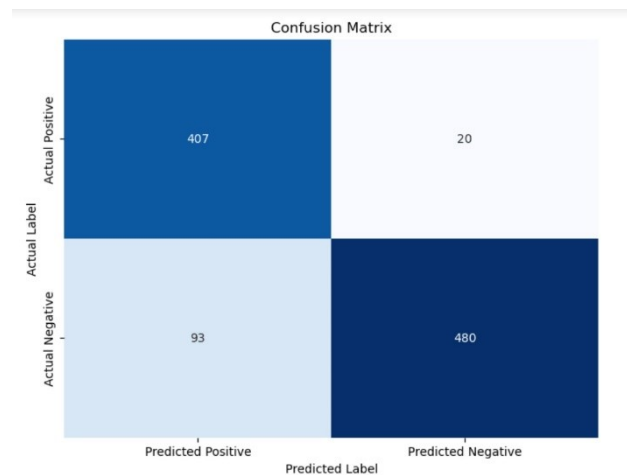


Figure .8: Confusion matrix

The figure.8 shows the model's classification performance. The model correctly predicted 407 positive cases (True Positives), while 20 positive cases were incorrectly classified as negative (False Negatives). It misclassified 93 negative cases as positive (False Positives), and correctly predicted 480 negative cases (True Negatives). The results suggest a strong model performance with low false positives and false negatives.

Further field tests, which involved real-time detection on various road types, validated the system's practical application. The pothole alert system, combined with real-time mapping to Firebase. The system's ability to process data in real-time ensures accurate pothole detection and mapping, enhancing road safety and aiding in proactive maintenance.



Figure .9: Pothole detected Images

Figure .9 shows images of potholes detected on the road, demonstrating the system's ability to identify hazards effectively. The experimental results confirm the reliability of the system, highlighting its potential to significantly reduce the risk of vehicle damage and accidents caused by potholes.

VI. CONCLUSION

The proposed pothole detection system, utilizing the YOLOv4-Tiny model on a Raspberry Pi, achieves high performance with a mean Average Precision (mAP) of 80.70%, surpassing models like YOLOv4 (77.70%) and YOLOv5s (74.80%) accuracy of 88.70%, precision of 81.4%, recall of 95.4%, and an F1-score of 87.8%. This standalone system, featuring an integrated camera, seamlessly connects to mobile internet and is easily mounted on the vehicle's dashboard. Once activated, it provides real-time pothole detection and location-based alerts to users via a Flutter mobile app. Data is efficiently stored and managed through Firebase Realtime Database and Storage, enabling road maintenance prioritization based on the collected data. By providing timely, reliable alerts, the system significantly enhances driver safety by helping users avoid potholes and prevent potential vehicle damage. Its seamless integration and solid performance ensure both the safety of the driver and the reliability of the data used for road maintenance. This system offers a practical, scalable solution that not only improves road safety but also supports more efficient road management for the future. Future enhancements could explore the addition of severity analysis for potholes, further improving the prioritization of road repairs. The proposed pothole detection system is efficient and practical, achieving a high mAP of 80.70% with strong accuracy metrics. It provides real-time alerts to drivers and stores data for road maintenance using a YOLOv4-Tiny model on a Raspberry Pi. This system effectively improves road safety and supports future enhancements like severity-based pothole analysis.

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